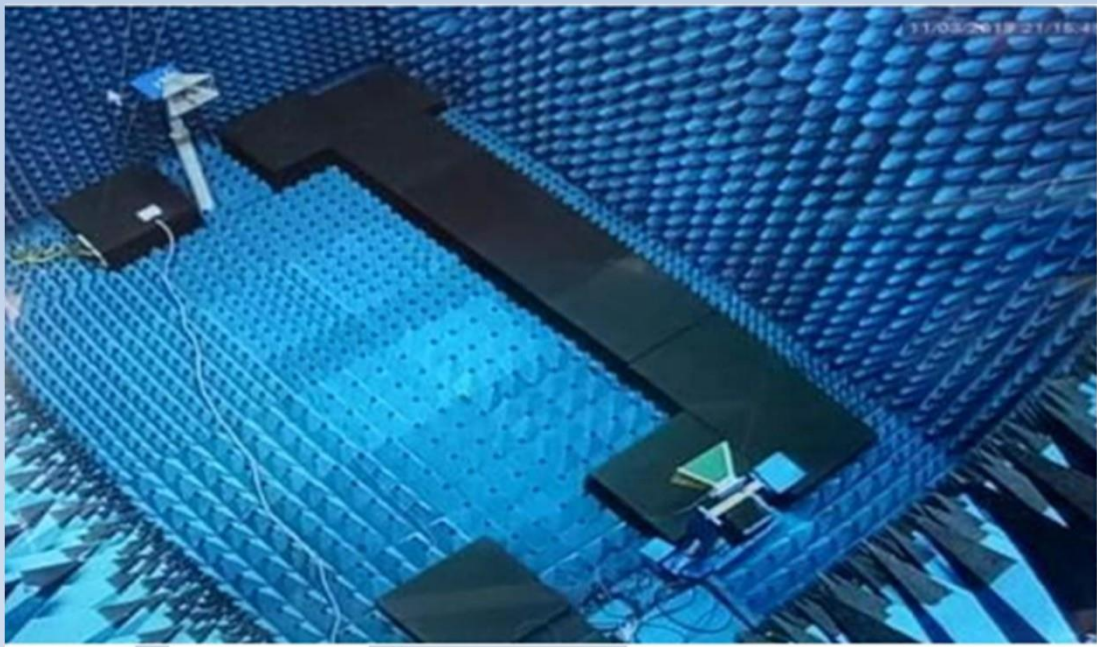


MICROLINE AUTOMATIC 3D ANTENNA MEASUREMENT SYSTEM

- **SUPPLY AND INSTALLATION OF POSITIONER SYSTEM**
- **POSITIONER SYSTEM OF BOTH TRANSMITTER AND RECEIVER**
- **SOFTWARE FOR MEASUREMENT**
- **ALL RELATED HARDWARE**



1. CUSTOMIZED DESIGN FOR 3D -2D PATTERN WITH PARAMETERS MEASUREMENT

CONSIST FOLLOWING HARDWARE & SOFTWARE

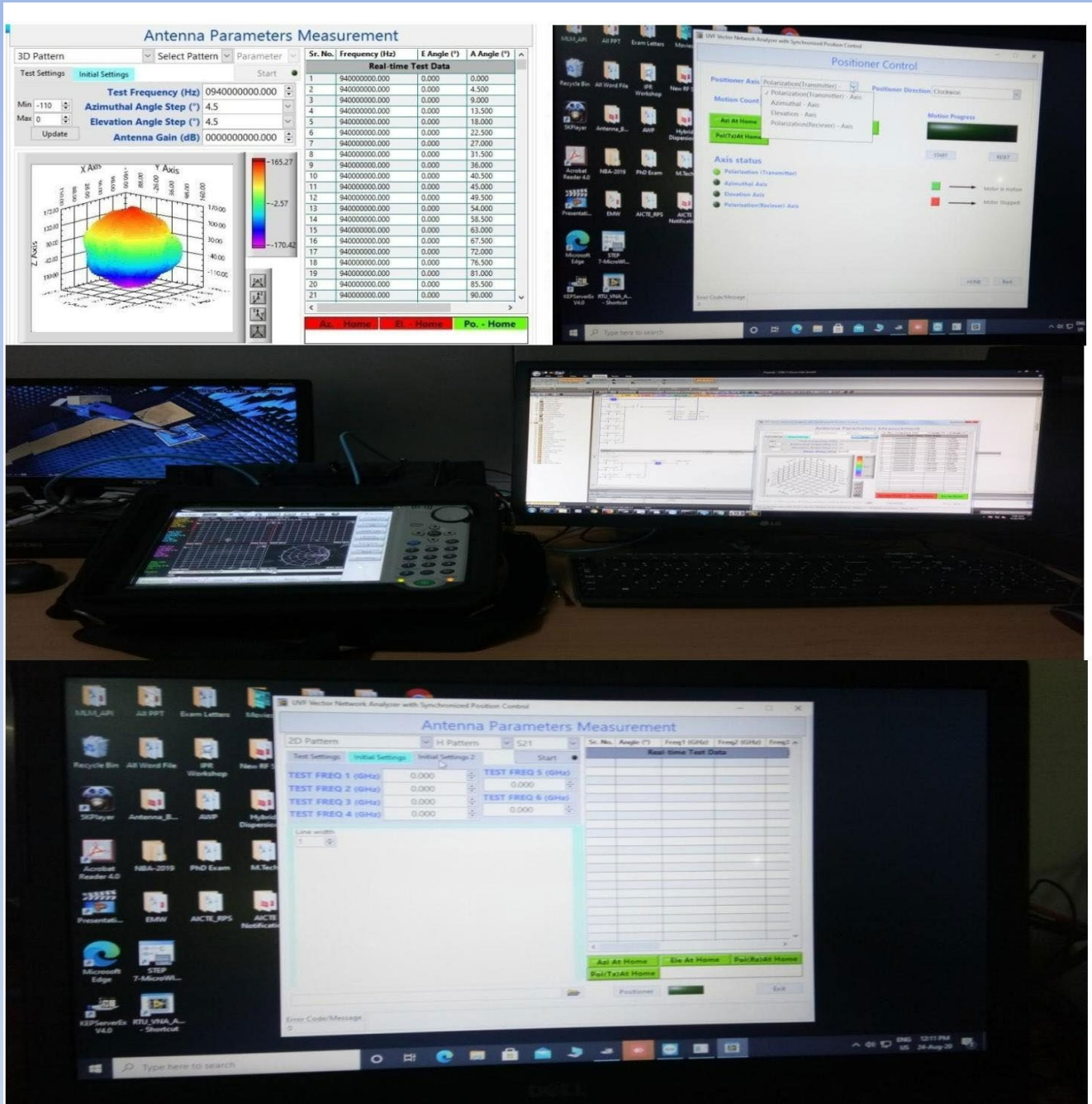
- 3 AXIS ROTARY TRANSMITTER POSITIONER SYSTEM
- 3 AXIS ROTARY RECEIVER POSITIONER SYSTEM
- AUTOMATIC RADIATION PATTERN MEASUREMENT SOFTWARE
- CONTROL SYSTEM HARDWARE
- MANUAL TO CONTROL AND DO MEASUREMENT

Automated Receiver and Transmitter System

Automatic 3 Axis receiver and Transmitter positioner system with similar movement of Elevation, Azimuthal rotations and Polarization

- i) Positioners would be of non-reflective materials
- ii) Motors and other parts would be covered with absorbers.
- iii) Interfacing with VNA and PC for automatic measurements through Software.
- iv) Minimum step angle would be 0.18 degree.
- v) Polarization and elevation movement form -90 degree to 90 degree and Azimuth movement from 0 to 360 degree.
- vi) After each measurement, positioner will return back to home position.
- vii) Multiple selected frequency option would be provided to get graph at the selected frequency points in single run of measurement. Six frequency data would be provided in Graph and as well as CSV data excel format.
- viii) Auto data saving and retrieving option will be provided.
- ix) Real time monitoring of graph and readings will be provided.
- x) Labview interfacing software will be provided along with PLC hardware controller. GUI will be Labview based.

- xi) Software will provide provision to have following measurements.
 - ✓ 2D Radiation pattern (E/ H)
 - ✓ 3D Radiation pattern (E/H)
 - ✓ Gain Measurement
 - ✓ Axial Ratio Measurement
 - ✓ S Parameter Measurement
- xii) After each measurement positioner system will return back to its home position.
- xiii) Both measurement control and graph plotting will be provided in the single screen.
- xiv) Positioner will be provided with mounting brackets to mount antennas.
- xv) Height of the Receiver and Transmitter positioner system can be varied manually with minimum scope of 150mm.



Above Picture showing – Pattern, Manual access to positioner, Interfacing pic from outside & Six frequency points data exe file pic.

Hardware Access Panel: Both positioners will rotate through controller and drive system which will be enclosed in one panel box. The panel will be placed Inside the chamber and covered with absorbers.

- ❖ Power to the panel will be provided from inside the chamber.
- ❖ LA cable will be taken through connector panel to the PC to have control and access from outside.

2. ABSORBERS

- Pyramidal Absorbers are flexible, light weight used in interference suppression applications
- Tested as per IEEE Standards
- Fire Retardant as per NRL 8093
- High performance from frequency band up to 40 GHz with attenuation of more than -20dB to -50 dB.
- Available in standard base size of 2' x 2' (610 mm x 610 mm).
- Temperature resistant
- Long performance warranty

Model MU 4

Model MU 4

Size : 610 x 610 x 100 mm (LWH)

Matrix : 12 x 12 = 144 pyramids Base Height : 25 mm

Pyramid Height : 75 mm

Reflectivity (Absorption level): -20 dB to -25 dB at 2 GHz

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